closer to real-time reasoning may be disabled or turned down. While this may limit the complexity at which an agent can interact, limited or no reasoning can improve response times and increase user enjoyment.

- *Customer service bot*—Again, because of the controlled nature of this environment, it's unlikely that assistants engaged directly with customers will need to perform complex or any form of reasoning.
- *Autonomous agent*—While reasoning is a strong component of autonomous agents, we still don't know how much reasoning is too much. As models such as Strawberry become available for agentic workflows, we can gauge at what point extensive reasoning may not be needed. This will surely be the case for well-defined autonomous agent workflows.
- *Collaborative workflows*—Again, applying reasoning creates an overhead in the LLM interaction. Extensive reasoning may provide benefits for some workflows, while other well-defined workflows may suffer. This may mean that these types of workflows will benefit from multiple agents—those with reasoning and those without.
- *Game AI*—Similar to other applications, heavy-reasoning applications may not be appropriate for most game AIs. Games will especially require LLM response times to be quick, and this will surely be the application of reasoning for general tactical agents. Of course, that doesn't preclude the use of other reasoning agents that may provide more strategic control.
- *Research*—Reasoning will likely be essential to any complex research task for several reasons. A good example is the application of the Strawberry model, which we've already seen in research done in mathematics and the sciences.

While we often consider reasoning in tandem with planning, there may be conditions where the level at which each is implemented may differ. In the next section we consider the agent pillar of evaluation of various applications.

11.5.3 Application of evaluation to agentic systems

Evaluation is the component of agentic/assistant systems that can guide how well the system performs. While we demonstrated incorporating